

COURSE NUMBER: CP1580

COURSE TITLE: Using Internet APIs

COURSE DESCRIPTION:

Including content and services from other providers is an important aspect of web development. In this course, the student will focus on how to use the APIs of other service providers to develop a web application. Topics will include an introduction to the APIs, and OAuth. Working with third party APIs and working with E-Commerce APIs. At the end of the course students will be able to create web applications that utilizes content provided through third party API's and will perform E-Commerce transactions utilizing payment processors APIs.

PREREQUISITES: CP1292 - JavaScript and Node JS

CO-REQUISITES: None

CREDIT VALUE: Three (3)

COURSE HOURS PER WEEK: Two (2)

LAB HOURS PER WEEK: Two (2)

REQUIRED TEXT: None

LEARNING RESOURCES:

[REST API Tutorial](#)

[Understanding and Using REST APIs](#)

[OAuth 2.0 Documentation](#)

[Using OAuth 2.0 to Access Google APIs](#)

[Weather API Documentation](#)

[The Movie Database API Documentation](#)

[The Spotify Web API Documentation](#)[PayPal API Documentation](#)

[Stripe API Documentation](#)

[Square API Documentation](#)

MAJOR TOPICS:

- 1.0 Introduction to APIs
- 2.0 Introduction to OAuth
- 3.0 Working with third party APIs
- 4.0 Working with E-Commerce APIs

LEARNING OBJECTIVES:

1.0 Introduction to APIs

- 1.1 Describe the SOAP protocol
- 1.2 Describe the REST Architectural style
- 1.3 Detail the benefits of the REST architecture style for web applications
- 1.4 Compare the REST verbs to the HTTP methods
 - 1.4.1 GET
 - 1.4.2 POST
 - 1.4.3 PUT
 - 1.4.4 DELETE
- 1.5 Use a REST API to
 - 1.5.1 Authenticate to a service
 - 1.5.2 Retrieve information from a service
 - 1.5.3 Send information to a service

2.0 Introduction of OAuth

- 2.1 Describe the OAuth Framework
- 2.2 Describe Scope in relation to OAuth
- 2.3 Detail the various Grant Types of Oath
- 2.4 Describe the OAuth Bearer Token
- 2.5 Use OAuth to gain access to a third party API

3.0 Working with third party APIs

- 3.1 Use a third party API to retrieve information from a service
- 3.2 Use a third party API to send information to a service

4.0 Working with E-Commerce APIs

- 4.1 Use a payment processors API to complete the following E-Commerce Transactions:
 - 4.1.1 Purchase an item
 - 4.1.2 Subscribe to a service
 - 4.1.3 Unsubscribe to a service
 - 4.1.4 Refund a transaction
- 4.2 Use the dashboard provided by the payment processor to confirm transactions

EVALUATION:

Assignments	30%
Test(s)	20%

Project 20%
Final Examination 30%

DATE DEVELOPED: September 2020

DATE REVIEWED:

REVISION NUMBER:

DATE REVISED:

Effective PeopleSoft Date: September 2021

Note to instructor: Check PIRS to ensure this outline is the most current version.